# Exercise 6 – Evaluation II

## Introduction to Machine Learning

*Hint: Useful libraries*

### R

```
# Consider the following libraries for this exercise sheet:

library(mlbench)
library(mlr3)
library(mlr3learners)
```

### Python

```
# Consider the following libraries for this exercise sheet:

# general
import numpy as np
import pandas as pd

# sklearn
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split
from sklearn.model_selection import RepeatedKFold
from sklearn.model_selection import RepeatedStratifiedKFold
```

## Exercise 1: Overfitting & underfitting

Learning goals

1. Discuss for given situations if they might lead to overfitting / underfitting
2. Judge whether over- and underfitting are properties of a learner or of a fixed model
3. Discuss to what extent over- and underfitting can be avoided

Assume a polynomial regression model with a continuous target variable $y$, a continuous, $p$-dimensional feature vector $\mathbf{x}$ and polynomials of degree $d$, i.e.,

$$f\left(\mathbf{x}^{(i)}\right) = \theta_0 + \sum_{j=1}^{p} \sum_{k=1}^{d} \theta_{j,k} (\mathbf{x}_j^{(i)})^k.$$

a) For each of the following situations, indicate whether we would generally expect the performance of a flexible polynomial learner (high $d$) to be better or worse than an inflexible one (low $d$). Justify your answer.

NB

We can only state tendencies here; performance strongly depends on the specific data situation.

i. The sample size $n$ is extremely large, and the number of features $p$ is small.

ii. The number of features $p$ is extremely large, and the number of observations $n$ is small.

iii. The true relationship between the features and the response is highly non-linear.

iv. The data could only be observed with a high level of noise.

b) Are overfitting and underfitting properties of a learner or of a fixed model? Explain your answer.

c) Should we aim to completely avoid both overfitting and underfitting?

## Exercise 2: Resampling in practice

Learning goals

1. Implement resampling procedures in R/Python
2. Understand how the choice of resampling strategy affects the quality of the GE estimator

a) Why would we apply resampling rather than a single holdout split?

b) Classify the `german_credit` data into solvent and insolvent debtors using logistic regression. Compute the training error w.r.t. MCE.

*Python Hint*

Read the already preprocessed file german_credit_for_py.csv

c) In order to evaluate your learner, compare the test MCE using

1. three times ten-fold cross validation (3x10-CV)
2. 10x3-CV
3. 3x10-CV with stratification for the feature `foreign_worker` to ensure equal representation in all folds
4. a single holdout split with 90% training data

*Hint*

**R**

You will need `rsmp`, `resample` and `aggregate`.

**Python**

You will need `RepeatedKFold`, `RepeatedStratifiedKFold` and `train_test_split`.

d) Discuss and compare your findings and compare them to the training error computed previously.

## Exercise 3: Computational cost of resampling

The whole exercise is only for lecture group B!

Learning goals

1. Determine how many models a resampling strategy fits
2. Reason about the computational cost of different resampling strategies
3. Relate computational cost to the quality of the GE estimate

Recall that a resampling strategy is a collection of $B$ train/test splits,

$$\mathcal{J} = \left((J_{\text{train},1}, J_{\text{test},1}), \dots, (J_{\text{train},B}, J_{\text{test},B})\right),$$

and that the learner is refitted on each of the $B$ training sets.

a) A data set has $n = 200$ observations. State how many models are fitted for each of the following strategies:

   i. 10-fold cross-validation (10-CV)
   ii. 5-times repeated 10-fold cross-validation (5x10-CV)
   iii. leave-one-out CV (LOO-CV)
   iv. subsampling with 25 iterations

*Hint*

The number of model fits equals the number of splits $B$.

b) Now give the number of model fits **in general**, i.e. as a formula, for a data set of size $n$ with $k$ folds, $r$ repetitions and $B$ iterations:

   i. a single holdout split
   ii. $k$-fold CV
   iii. $r$-times repeated $k$-fold CV
   iv. LOO-CV
   v. subsampling or bootstrap with $B$ iterations

c) Explain why LOO-CV becomes computationally prohibitive for large $n$, and name a resampling strategy that achieves a similarly low (pessimistic) bias at much lower cost.

d) Both 3-times repeated 10-fold CV (3x10-CV) and 10-times repeated 3-fold CV (10x3-CV) fit exactly 30 models, so their computational cost is roughly the same. Does this mean that both strategies provide GE estimates of the same quality? Explain your answer.

## Exercise 4: Variance and independence of GE estimators

The whole exercise is only for lecture group A!

Learning goals

1. Derive mean and variance of the holdout GE estimator on a fresh i.i.d. test set
2. Understand why the per-fold errors of cross-validation are not independent

Consider a fixed model $\hat{f}$ that has been fitted on some training data and is now held fixed. We estimate its generalization error on a test set $\mathcal{D}_{\text{test}}$ of $n_{\text{test}}$ observations drawn i.i.d. from $\mathbb{P}_{xy}$, independently of the training data:

$$\widehat{\text{GE}}\left(\hat{f}\right) = \frac{1}{n_{\text{test}}} \sum_{(\mathbf{x},y)\in\mathcal{D}_{\text{test}}} L\left(y, \hat{f}(\mathbf{x})\right).$$

a) Show that $\widehat{\text{GE}}\left(\hat{f}\right)$ is an unbiased estimator of the theoretical generalization error $\text{GE}\left(\hat{f}\right) = \mathbb{E}\left[L\left(y, \hat{f}(\mathbf{x})\right)\right]$ of the fixed model $\hat{f}$.

b) Show that $\mathsf{Var}\left[\widehat{\text{GE}}\left(\hat{f}\right)\right] = \frac{1}{n_{\text{test}}}\mathsf{Var}\left[L\left(y, \hat{f}(\mathbf{x})\right)\right]$ and interpret how the estimator's precision depends on the test set size.

c) In $k$-fold cross-validation we obtain $k$ per-fold estimates $\widehat{\text{GE}}^{(1)}, \ldots, \widehat{\text{GE}}^{(k)}$ and report their mean. One might be tempted to treat these $k$ numbers as $k$ independent measurements – e.g., to report their standard deviation or run a $t$-test to compare two learners. Explain why the $\widehat{\text{GE}}^{(j)}$ are **not** independent, and what this implies for such a variance estimate.